Monte Carlo simulation

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1. Snake Eyes:
    a. 100 hands vs 100,000 hand of rolling 24 times:
    The result for checkPascal(100, 24) was 0.55 and
    The result for checkPascal(100000, 24) was 0.49219
    They are not exactly the same. The 100,000 hands of rolling 2 dies 24 was closer to the expected value.

    b. Calculation of the probability of getting snake eyes in 10 rolls:
    So first the probability of not rolling snake eye in 10 rolls is (35/36)^10 = 0.754
    So now the probability of rolling snake eyes in 10 rolls is (1 - 0.754) = 0.246

    c. When I ran checkPascal(10000, 10)
    I got Probability of winning = 0.2505.
    The error difference between this and my expected result is:
    0.2505 - 0.246 = 0.0045

2. Whether playing the Pass Line or the Don't Pass Line has a higher probability of winning
    a. Running the code to 20 hands with 10 games.
    First Run:
    The 1st run of results of crapsSim(20, 10) are:
    Pass: Mean ROI = 5.0% Std. Dev. = 18.0278%
    Don't pass: Mean ROI = -7.0% Std Dev = 18.4662%

    Second Run:
    The 2nd run of results of crapsSim(20, 10) are:
    Pass: Mean ROI = -13.0% Std. Dev. = 16.1555%
    Don't pass: Mean ROI = 10.5% Std Dev = 15.7242%

    These results are not the same.

    b. Increasing the number of hands, and how many hands we need to be confident that the bet is either good or bad:
    I increased the number of hands to 10000 crapsSim(10000, 10) and got these results:
    Pass: Mean ROI = -1.594% Std. Dev. = 0.7597%
    Don't pass: Mean ROI = -1.136% Std Dev = 0.8848%

    I increased the number again to 100000 crapsSim(100000, 10) and got these results:
    Pass: Mean ROI = -1.4968% Std. Dev. = 0.1801%
    Don't pass: Mean ROI = -1.2944% Std Dev = 0.1609%

It seems like 10000 hands is a good number of hands to be confident that the bet is good. I ran crapsSim(10000, 10) multiple times and the signs didn't change between the runs. Increased it to 100,000 hands, it had similar behavior.

c. ROI results, which is higher probability, which is better?
The ROI I got for Pass Line was -1.4968% and for Don't Pass Line -1.2944%
The return on investment for the Pass was lower than the Don't Pass Line. Technically the Don't Pass Line is a better bet, but I would say neither are very good bets.

d. Cheating:
I changed rollDie() to rollDie2() and got these results after I ran crapsSim(10000, 10):
Pass: Mean ROI = 7.122% Std. Dev. = 0.845%
Don't pass: Mean ROI = -9.86% Std Dev = 0.8215%

The results indicate that the Pass Line has a higher probability, so it's a "good" bet.

e. A biased die that gives a higher ROI than rollDie2() for the Pass Line:

I created a die that favors higher numbers:

[5,5,5,6,6,6,6,4,4,3,2,1]

After running crapsSim(10000, 10), I got:

Pass: Mean ROI = 13.074%

Don't pass: Mean ROI = -24.232%

The Pass Line ROI increased compared to rollDie2()

f. A biased die that is a good bet for the Don't Pass Line:

I created a biased die that favors lower numbers:

[1,1,1,2,2,3,3,4,4,5,6,6]

This increases the probability of rolling 2 and 3 on the first throw.

After running crapsSim(10000, 10), I got:

Pass: Mean ROI = −10.74%
Don't Pass: Mean ROI = 8.013%

Since the Don't Pass Line has a positive return on investment, this biased die makes it a good bet.

3. Crapless Craps

```
def craplessSim(handsPerGame, numGames, dieFunc):
    games = []

    for t in range(numGames):
        g = CraplessGame(dieFunc)
        for i in range(handsPerGame):
            g.playHand()
        games.append(g)

    ROIs = []
    for g in games:
        wins, losses = g.results()
        ROIs.append((wins - losses) / float(handsPerGame))

    meanROI = str(round(100 * (sum(ROIs) / numGames), 4)) + '%'
    sigma = str(round(100 * stdDev(ROIs), 4)) + '%'
    print('Crapless Craps:', 'Mean ROI =', meanROI, 'Std. Dev. =', sigma)

craplessSim(10000, 10, rollDie)
craplessSim(10000, 10, rollDie2)
craplessSim(10000, 10, rollDie4)
```

Crapless Craps: Mean ROI = -5.86% Std. Dev. = 1.0391%
Crapless Craps: Mean ROI = -1.998% Std. Dev. = 0.9237%
Crapless Craps: Mean ROI = -6.04% Std. Dev. = 0.8246%

a. The odds become worse with original dice
b. The odds become better with biased dice (but depends on the biased dice)